Revision One-Pagers

42 Topics | 5 Subjects | 3rd Year MBBS

Everything on ONE page per topic!

Community Medicine (PSM) | Ophthalmology | ENT | Forensic Medicine | Clinical Postings

Last-minute revision made easy!

Each page = 1 complete topic compressed. Slide through before your exam!

42 TOPICS

EXAM READY!

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42 Revision One-Pagers

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Epidemiology

DEFINITION

Study of distribution and determinants of disease frequency in human populations. — John Last

KEY FACTS

1. Epidemiological Triad: Agent + Host + Environment
2. Observational: Cross-sectional (prevalence), Case-control (OR, backward), Cohort (RR, forward)
3. RCT = gold standard interventional study
4. Incidence = new cases / pop at risk x time; Prevalence = all cases / total pop
5. Prevalence = Incidence x Duration
6. Relative Risk (RR) = cohort; Odds Ratio (OR) = case-control; OR = ad/bc from 2x2 table
7. Bradford Hill: 9 criteria, TEMPORALITY = only essential one
8. Bias types: Selection, Information, Recall, Observer, Hawthorne, Lead-time, Length-time
9. Confounding: associated with both exposure & outcome, NOT on causal pathway; Control by randomization, matching, stratification
10. Levels of Prevention: Primordial → Primary → Secondary → Tertiary

Study Designs

Design	Direction	Measure	Key Feature
Cross-sectional	Snapshot	Prevalence	Quick & cheap
Case-control	Effect→Cause	Odds Ratio	Good for rare diseases
Cohort	Cause→Effect	Relative Risk	Good for rare exposures
RCT	Assigned	RR / NNT	Gold standard causation
Ecological	Group-level	Correlations	Ecological fallacy risk

Me seeing '2x2 table' in the exam

I was BORN for this moment

Memory Tricks!

Case-control = BACKWARD = Odds Ratio | Cohort = FORWARD = Relative Risk

TEMPORALITY is the ONLY essential Bradford Hill criterion!

OR = ad/bc (criss-cross from 2x2 table)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Best study for rare disease?	Case-control
Q Best study for causation?	RCT
Q Prevalence = ?	Incidence x Duration
Q Hawthorne effect?	Behavior changes when observed
Q NNT formula?	1 / ARR

DEFINITION

Science of collecting, summarizing, analyzing & interpreting data in medicine and public health.

KEY FACTS

1. Data types: Nominal, Ordinal, Interval, Ratio (NOIR)
2. Mean affected by outliers; Median robust; Mode for categorical
3. Mode = 3(Median) - 2(Mean) — empirical formula
4. Normal distribution: 68-95-99.7 rule (±1SD, ±2SD, ±3SD)
5. SD = variability in sample; SE = SD/√n = precision of mean
6. Sensitivity = a/(a+c); Specificity = d/(b+d); PPV = a/(a+b); NPV = d/(c+d)
7. SnNOut: High Sensitivity + Negative = rules OUT disease
8. SpPIn: High Specificity + Positive = rules IN disease
9. $p < 0.05$ = statistically significant; 95% CI = Mean ± 1.96 x SE
10. Type I (α) = false positive (0.05); Type II (β) = false negative (0.20)

Statistical Tests

Scenario	Parametric	Non-parametric
2 groups, paired	Paired t-test	Wilcoxon signed-rank
2 groups, unpaired	Unpaired t-test	Mann-Whitney U
3+ groups	ANOVA	Kruskal-Wallis
Categorical data	—	Chi-square (X^2)
Correlation	Pearson r	Spearman rho

Stats professor: '$p < 0.05$'

Me: significant enough to pass??

Memory Tricks!

68-95-99.7 Rule — most drawn diagram in PSM!

SnNOut / SpPIn — sensitivity & specificity clinical rules

Chi-square: $X^2 = \Sigma(O-E)^2/E$; df = (rows-1)×(cols-1)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Effect of prevalence on PPV?	Higher prev → higher PPV
Q r ranges from?	-1 to +1
Q Coefficient of determination?	r^2 (variance explained)
Q Power of test?	1 - β
Q Yates correction when?	Expected value < 5

Screening Tests

DEFINITION

Presumptive identification of unrecognized disease by tests applied rapidly and economically to healthy populations.

KEY FACTS

1. Wilson & Jungner Criteria (WHO 1968): 10 principles for screening programs
2. Screening = rapid, cheap, applied to healthy; Diagnostic = confirms, applied to suspects
3. 2x2 table: TP (a), FP (b), FN (c), TN (d)
4. Sensitivity = a/(a+c); Specificity = d/(b+d)
5. PPV varies with prevalence; Sensitivity/Specificity are fixed test properties
6. ROC Curve: Sensitivity (Y) vs 1-Specificity (X); AUC = accuracy (1.0 perfect, 0.5 useless)
7. LR+ > 10 = strong for disease; LR- < 0.1 = strong against
8. Screening biases: Lead-time, Length-time, Selection, Overdiagnosis

Screening vs Diagnostic Test

Feature	Screening	Diagnostic
Applied to	Apparently healthy	Suspected cases
Cost	Low	High
Speed	Rapid	Time-consuming
Purpose	Detect early disease	Confirm diagnosis
Sensitivity	HIGH preferred	HIGH + specific

ELISA says positive

Western Blot: hold my beer

Memory Tricks!

ELISA (high Sensitivity) ⊠ SnNOut; Western Blot (high Specificity) ⊠ SpPIn

Lead-time bias: early detection looks like longer survival, but death unchanged

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Gold standard cervical CA screening?	HPV DNA / Pap smear
Q PKU screening?	Guthrie test (heel prick)
Q Neonatal hypothyroidism screen?	TSH
Q Higher prevalence effect?	PPV ⊠, NPV ⊠
Q Wilson & Jungner year?	WHO 1968

National Health Programs

DEFINITION

Organized government programs to control/eliminate specific diseases and promote health at national level.

KEY FACTS

1. NTEP (ex-RNTCP): TB elimination by 2025; DOTS mandatory; 2mo HRZE + 4mo HRE daily
2. NACP: HIV/AIDS 5 phases; end AIDS by 2030
3. UIP (1985): 12 VPDs; Pentavalent = DPT+HepB+Hib
4. NVBDCP: Malaria (ACT), Dengue, Filariasis (DEC+Albendazole MDA), Kala-azar
5. NLEP: Leprosy elimination <1/10,000 achieved 2005; MDT: PB 6mo, MB 12mo
6. Pulse Polio 1995: India polio-free 2011; certified 2014
7. NHM (2013) = NRHM + NUHM; ASHA, JSY, JSSK key schemes
8. Ayushman Bharat: HWC + PM-JAY (Rs 5 lakh/family/yr)

Major NHPs at a Glance

Programme	Year	Key Goal
NMCP (Malaria)	1953	First NHP in India
NLEP (Leprosy)	1983	MDT; elimination 2005
UIP	1985	12 VPDs; Pentavalent
RNTCP/NTEP	1997/2020	TB elimination 2025; DOTS
NACP (HIV)	1992	End AIDS 2030; 5 phases
NHM	2013	NRHM + NUHM umbrella

Examiner: name 10 NHPs

My brain: DOTS... that's all I got

Memory Tricks!

DOTS = Directly Observed Treatment Short-course

MDT: PB = Dapsone + Rifampicin (6mo); MB = + Clofazimine (12mo)

ASHA = Accredited Social Health Activist (NRHM key worker)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q First NHP in India?	National Malaria Control (1953)
Q PM-JAY coverage?	Rs 5 lakh/family/year
Q Pentavalent covers?	DPT + HepB + Hib
Q India polio-free year?	2011 (certified 2014)
Q DOTS stands for?	Directly Observed Tx Short-course

Environmental Health

DEFINITION

Health maintenance through safe water, sanitation, housing, air quality, and waste management.

KEY FACTS

1. Boiling = most reliable household water purification
2. Slow sand filter: 100-200 L/m²/hr, Schmutzdecke (bio layer), 98-99% removal, best rural
3. Rapid sand filter: 3000-6000 L/m²/hr, mechanical, needs disinfection, urban use
4. Standard free residual chlorine = 0.5 mg/L after 30 min contact
5. Break-point chlorination: 3 phases ⊠ free Cl appears after break-point
6. Horrock's apparatus: determines chlorine dose for wells
7. BOD < 2 mg/L = clean water; COD always > BOD
8. Sewage treatment: Preliminary ⊠ Primary ⊠ Secondary (biological) ⊠ Tertiary ⊠ Disinfection
9. Swachh Bharat Mission 2014: ODF goal 2019; Jal Jeevan Mission: Har Ghar Jal by 2024

Slow vs Rapid Sand Filter

Feature	Slow Sand	Rapid Sand
Rate	100-200 L/m²/hr	3000-6000 L/m²/hr
Mechanism	Biological (Schmutzdecke)	Mechanical only
Bacterial removal	98-99%	90-95%
Post-disinfection	Not essential	Essential
Best for	Rural areas	Urban areas

Boil water before drinking!

Me after learning Schmutzdecke exists

Memory Tricks!

0.5 mg/L free chlorine = gold standard residual

OT test = Ortho-tolidine (yellow); DPD = most accurate; Starch-iodide (blue)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Most reliable purification?	Boiling
Q Most widely used disinfectant?	Chlorine
Q Best biological filter?	Slow sand (Schmutzdecke)
Q BOD of clean water?	< 2 mg/L
Q Standard residual chlorine?	0.5 mg/L free Cl

DEFINITION

Disease arising out of or during employment, directly related to workplace exposure.

KEY FACTS

1. Hazards: Physical, Chemical, Biological, Mechanical, Psychosocial
2. Prevention hierarchy: Elimination ⊠ Substitution ⊠ Engineering ⊠ Admin ⊠ PPE (least effective)
3. Silicosis = most common pneumoconiosis (SiO_2, egg-shell calcification, TB risk)
4. Asbestosis = most fibrogenic (mesothelioma, asbestos bodies)
5. Byssinosis = cotton dust, Monday fever (symptoms return Monday after weekend off)
6. Lead: Burton's line (blue gum), wrist drop, colic, basophilic stippling
7. NIHL: ⊠85 dB threshold; 4 kHz dip = earliest finding; doubling rule (3 dB = halve time)
8. Heat stroke: temp >40.6°C, NO sweating — EMERGENCY
9. Factories Act 1948: max 48 hrs/week, 9 hrs/day

Pneumoconioses

Disease	Dust	Key Feature
Silicosis	SiO_2 (quartz)	Egg-shell calcification, TB risk
Asbestosis	Asbestos	Mesothelioma, most fibrogenic
Byssinosis	Cotton dust	Monday fever, reversible early
Bagassosis	Sugarcane	Hypersensitivity pneumonitis
Anthracosis	Coal dust	Coal macules, CWP

Monday morning at the cotton mill

Byssinosis: 'did you miss me?'

Memory Tricks!

Lead ABCD: Anaemia, Burton's line, Colic, Drop (wrist/foot)

Silicosis = most COMMON; Asbestosis = most FIBROGENIC

Permissible noise = 85 dB (OSHA); doubling rule: +3 dB = halve exposure time

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Most common pneumoconiosis?	Silicosis
Q Most fibrogenic?	Asbestosis
Q Monday fever?	Byssinosis
Q Burton's line?	Lead poisoning
Q Wrist drop occupation?	Lead poisoning

DEFINITION

Balance of macro & micronutrients needed for body growth, development, and health maintenance.

KEY FACTS

1. Calories: Carbs 4 kcal/g, Protein 4 kcal/g, Fat 9 kcal/g
2. Mutual supplementation: Cereal + Pulse = complete protein (dal-chawal)
3. BMI: Normal 18.5-24.9 (WHO), 18.5-22.9 (Asian cut-off)
4. Kwashiorkor: Protein deficiency, 1-3yr, OEDEMA hallmark, flag sign hair, flaky paint dermatosis
5. Marasmus: Total calorie+protein, <1yr, NO oedema, severe wasting, monkey face, good appetite
6. Vitamin A: Night blindness ⊠ Bitot spots ⊠ Keratomalacia; Dose: 1L IU at 9mo, 2L IU 6-monthly
7. Pellagra (B3): 3 Ds = Dermatitis, Diarrhea, Dementia
8. ICDS (1975): World's largest child programme; 6 services at AWC

Kwashiorkor vs Marasmus

Feature	Kwashiorkor	Marasmus
Deficiency	Protein	Total calories + protein
Age	1-3 years	< 1 year
Oedema	PRESENT (hallmark)	Absent
Wasting	Mild (hidden by oedema)	Severe (skin & bone)
Appetite	Poor	Good (ravenous)
Face	Moon face	Monkey/old man face
Hair	Flag sign, sparse	Thin, dry

Kwashiorkor vs Marasmus?

One is puffy, one is bony. DONE.

Memory Tricks!

K = Kwashiorkor = Keep Oedema (hallmark)

M = Marasmus = Muscles Melted (severe wasting, no oedema)

Pellagra = 3 Ds: Dermatitis, Diarrhea, Dementia

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Oedema + flaky paint?	Kwashiorkor
Q Monkey face?	Marasmus
Q Bitot's spots?	Vitamin A deficiency
Q 3 Ds?	Pellagra (Niacin/B3)
Q Flag sign of hair?	Kwashiorkor

DEMOGRAPHY

DEFINITION

Scientific study of human populations — size, structure, distribution, and changes (births, deaths, migration).

KEY FACTS

1. Demographic Transition: Stage 1 (high BR/DR) ⊠ Stage 2 (India: high BR, declining DR) ⊠ Stage 3 ⊠ Stage 4 (low BR/DR)
2. India Census 2011: 121 crore, sex ratio 943, child ratio 914, literacy 74%
3. SRS 2020: CBR 19.5, CDR 6.0, IMR 28, NMR 20, MMR 97/lakh, TFR 2.0
4. TFR 2.1 = replacement level; India TFR 2.0 (below replacement!)
5. IMR = Deaths <1yr / Live births x 1000; best single health indicator
6. Population Pyramid: Expansive (India), Constrictive (Japan), Stationary (Sweden)
7. Doubling Time = 70 / Growth Rate (%)
8. NPP 2000: Target TFR 2.1 by 2010; stable population by 2045
9. Most popular FP method India = Tubectomy (36%)

Demographic Transition Stages

Stage	Birth Rate	Death Rate	Growth	Example
1	High	High	Low/zero	Pre-industrial
2	High	Declining	Rapid	India (current)
3	Declining	Low	Slowing	Brazil, Mexico
4	Low	Low	Zero/negative	Japan, Sweden

India TFR dropped below 2.1!

Demographic dividend goes brrr

Memory Tricks!

India sex ratio 943 overall, 914 child (declining!)

MMR denominator = per LAKH (not per 1000!)

India = first country with national FP programme (1952)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Best health indicator?	IMR
Q India IMR (SRS 2020)?	28/1000
Q Replacement TFR?	2.1
Q Most popular FP India?	Tubectomy
Q First country with FP programme?	India (1952)

Health Indicators

DEFINITION

Variable measured directly to reflect the state of health of a community or population.

KEY FACTS

1. IMR = Deaths <1yr / Live births x 1000 = best overall health indicator
2. NMR contributes ~70% of IMR (focus on first 28 days!)
3. MMR = Maternal deaths / Live births x 1,00,000 (per LAKH); India 97/lakh
4. #1 cause maternal death = PPH (postpartum hemorrhage)
5. DALY = YLL + YLD (1 healthy year LOST — higher = worse)
6. QALY = Years x Quality weight (1 healthy year GAINED — higher = better)
7. HDI (UNDP) = Health + Education + Income; India 0.644, Rank 134
8. Crude rates misleading by age; use Standardized rates for fair comparison

DALY vs QALY

Feature	DALY	QALY
Measures	Health LOST	Health GAINED
Higher =	Worse	Better
Components	YLL + YLD	Years x Quality (0-1)
Used for	Disease burden (WHO)	Cost-effectiveness
1 unit =	1 healthy year lost	1 healthy year gained

MMR denominator is per LAKH

Not per 1000... I've been wrong ALL year

Memory Tricks!

DALY = health LOST (bad); QALY = health GAINED (good)

NMR = 70% of IMR (biggest impact in first 28 days)

HDI = LE + Education + Income

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Best single health indicator?	IMR
Q MMR denominator?	Per 1,00,000 live births
Q #1 maternal death cause?	Hemorrhage (PPH)
Q DALY formula?	YLL + YLD
Q India HDI rank?	134/193

Health System in India

DEFINITION

Organized three-tier system providing health services at primary, secondary, and tertiary levels.

KEY FACTS

1. Sub-Centre: 5000 pop, no beds, ANM + male HW, most peripheral facility
2. PHC: 30,000 pop, 6 beds, MO + staff, first contact with qualified doctor (rural)
3. CHC: 1,20,000 pop, 30 beds, 4 specialists (Surgery, Medicine, OBG, Paediatrics), FRU
4. District Hospital: 75-500 beds, full specialist services, highest referral
5. Each PHC covers 6 SCs; each CHC covers 4 PHCs
6. HWC under Ayushman Bharat: 1,50,000 target, CPHC (12 service packages)
7. ASHA: 1 per 1000 pop, incentive-based (NOT salary), link community ⊠ health system
8. Govt health expenditure ~1.35% GDP (target 2.5%); OOPE ~47%

Health Facility Comparison

Facility	Population	Beds	Staff
Sub-Centre	5,000	0	ANM + Male HW
PHC	30,000	6	MO + Staff Nurse
CHC	1,20,000	30	4 Specialists
District Hospital	Per district	75-500	Full specialists

SC ⊠ PHC ⊠ CHC ⊠ DH

Me explaining Indian healthcare pyramid

Memory Tricks!

SC ⊠ PHC ⊠ CHC ⊠ DH (pyramid)

Population: 5K ⊠ 30K ⊠ 1.2L ⊠ District

ASHA ⊠ ANM: ASHA = volunteer (incentive), ANM = professional (salary)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Most peripheral facility?	Sub-Centre
Q First qualified doctor (rural)?	PHC
Q PHC population (plains)?	30,000
Q CHC beds?	30
Q 4 specialists at CHC?	Surg, Med, OBG, Paeds

DEFINITION

Three coats: Fibrous (sclera+cornea), Vascular/Uveal (iris+ciliary body+choroid), Nervous (retina with 10 layers).

KEY FACTS

1. Cornea: 5 layers (EBSDE: Epithelium, Bowman's, Stroma 90%, Descemet's, Endothelium)
2. Total refractive power: 58-60D; Cornea +43D (2/3), Lens +15-20D (1/3)
3. Central retinal artery = END artery (no anastomoses)
4. CN III: SR, IR, MR, IO, LPS + parasympathetic | CN IV: SO | CN VI: LR
5. Annulus of Zinn: common tendinous ring, origin of 4 recti muscles
6. Lens: avascular, grows throughout life, capsule = longest basement membrane
7. Limbus: sclera-cornea junction; Schlemm's canal drains aqueous
8. AC depth ~2.5mm; Aqueous 0.25ml; Vitreous 4ml (99% water)

Cranial Nerves of Eye

Nerve	Muscles	Key Feature
CN III	SR, IR, MR, IO, LPS	Also parasympathetic
CN IV	Superior Oblique	Longest, thinnest CN
CN VI	Lateral Rectus	Longest intracranial course
CN II	Optic nerve	Sensory, part of CNS
CN V1	Sensory to cornea	Corneal reflex afferent

LR6 SO4 Rest3

If you forget this, there's no hope

Memory Tricks!

LR6 SO4 Rest3 (LR=CNVI, SO=CNIV, all others=CNIII)

Cornea layers: EBSDE (Epi, Bowman's, Stroma, Descemet's, Endo)

Retina 10 layers: I Play Extremely Good Innings, I Never Gave Losers Runs

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Thinnest orbit wall?	Medial (lamina papyracea)
Q End artery of eye?	Central retinal artery
Q 90% of corneal thickness?	Stroma
Q Highest refractive power?	Cornea (+43D)
Q NOT from annulus of Zinn?	Inferior oblique

Physiology of Vision

DEFINITION

Visual pathway: Retina ⊠ Optic nerve ⊠ Chiasma ⊠ LGB ⊠ Visual cortex (Area 17). Nasal fibers cross, temporal uncross.

KEY FACTS

1. Rods: 120M, scotopic (dim light), high sensitivity | Cones: 6M, photopic (bright + color), high acuity
2. Fovea: cones only, highest acuity, no blood vessels
3. Pupillary light reflex: Afferent CN II ⊠ Pretectal ⊠ Bilateral EW ⊠ Efferent CN III
4. Accommodation = convergence + ciliary contraction + pupil constriction (CN III)
5. Argyll Robertson pupil: accommodates but NO light reflex (neurosyphilis)
6. Marcus Gunn (RAPD): optic nerve lesion ⊠ paradoxical dilation on swinging flashlight
7. Dark adaptation: Cones 7-8 min, Rods 30-40 min; Rhodopsin regenerates in dark
8. Color vision: Ishihara plates (screening); Deuteranopia (green) most common deficiency

Rods vs Cones

Feature	Rods	Cones
Number	120 million	6 million
Location	Periphery	Fovea concentrated
Function	Dim light (scotopic)	Bright + color (photopic)
Pigment	Rhodopsin (Vit A)	Iodopsin (3 types)
Acuity	Low	High (1:1 at fovea)

Argyll Robertson pupil

Accommodates but won't react… mood

Memory Tricks!

Argyll Robertson = ARP: Accommodation Reflex Present, light absent ('prostitute's pupil')

RAPD = Relative Afferent Pupillary Defect = optic nerve lesion

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Pituitary tumor field defect?	Bitemporal hemianopia
Q Most common color blindness?	Deuteranopia (green)
Q Color vision screening?	Ishihara plates
Q Pupil accommodates but no light?	Argyll Robertson
Q Macular sparing in?	Occipital cortex lesion

Refraction Errors

DEFINITION

Optical state where parallel rays fail to focus on retina: myopia, hypermetropia, astigmatism, presbyopia.

KEY FACTS

1. Myopia: image IN FRONT of retina; concave (minus) lens; eye too long/powerful
2. Hypermetropia: image BEHIND retina; convex (plus) lens; eye too short/weak
3. 1mm axial length change = 3D refraction change
4. Presbyopia: loss of accommodation >40yr; +1D at 40, +1.5 at 45, +2 at 50, +3 at 60
5. Astigmatism: WTR (vertical, young) vs ATR (horizontal, old)
6. Pathological myopia >6D: staphyloma (MC), Foster-Fuchs spot, RD risk
7. LASIK: >18yr, stable >1yr, CCT >500µm, up to -10D
8. Retinoscopy: WITH movement = Hypermetropia; AGAINST = Myopia

Myopia vs Hypermetropia

Feature	Myopia	Hypermetropia
Image falls	In FRONT	BEHIND retina
Eye is	Too long/powerful	Too short/weak
Lens Rx	Concave (minus)	Convex (plus)
Vision affected	Distance	Near (first)
AC depth	Deep	Shallow
Angle closure risk	LOW	HIGH

Patient: 'I can't see far'

Me: concave lens go brr

Memory Tricks!

WITH movement = Hypermetropia (use +); AGAINST = Myopia (use -)

WTR = Young (standing tall); ATR = Old (lying down)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q #2 cause visual impairment India?	Refractive errors (19.7%)
Q Posterior staphyloma?	Degenerative myopia
Q Spasm of accommodation mimics?	Myopia (pseudo-myopia)
Q Best objective test children?	Retinoscopy under cycloplegia
Q Presbyopia starts at?	Age 40

Cataract

DEFINITION

Any opacity in the crystalline lens or its capsule, irrespective of effect on vision. #1 cause of blindness in India (62.6%).

KEY FACTS

1. Senile (>90%): Nuclear (myopic shift, brown), Cortical (spoke-like, glare), PSC (rapid, near loss first)
2. 4 Stages: Immature (iris shadow+) ⊠ Mature (iris shadow-, pearly white) ⊠ Hypermature (shrunken) ⊠ Morgagnian (liquefied cortex)
3. Second sight = nuclear sclerosis ⊠ myopic shift ⊠ reads without glasses (temporary!)
4. SICS vs Phaco vs ECCE: SICS (6-7mm, sutureless, best India), Phaco (2.8mm, gold standard global), ECCE (10-12mm, sutures)
5. SICS = surgery of choice in Indian mass camps (low cost, no machine)
6. IOL: SRK Formula P = A - 2.5L - 0.9K
7. PCO = most common LATE complication; Rx = Nd:YAG capsulotomy
8. Phacomorphic (swollen lens ⊠ angle closure) vs Phacolytic (protein leak ⊠ OAG)
9. NPCBVI (1976): Target 0.3% blindness; CSR ~6000/million/yr

SICS vs Phaco vs ECCE

Feature	SICS	Phaco	ECCE
Incision	6-7mm	2.8-3.2mm	10-12mm
Sutures	No	No	Yes
Recovery	1-2 wks	<1 wk	6-8 wks
Cost	Low	High	Moderate
Best for	India (mass)	Soft nuclei	No phaco/SICS

Second sight = reads without glasses!

Plot twist: cataract still growing

Memory Tricks!

Iris shadow test: Present = Immature; Absent = Mature

Phacomorphic = swollen lens (angle closure); Phacolytic = leaky lens (OAG)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q #1 cause blindness India?	Cataract (62.6%)
Q Oil droplet cataract?	Galactosemia
Q Sunflower cataract?	Wilson disease
Q Most common late complication?	PCO (Nd:YAG Rx)
Q Rosette cataract?	Trauma (concussion)

Glaucoma

DEFINITION

Progressive optic neuropathy with characteristic disc & field changes. IOP is major risk factor but NOT the definition.

KEY FACTS

1. Normal IOP: 10-21 mmHg; highest at 4-6 AM
2. Aqueous drainage: 90% trabecular (TM ⊠ Schlemm ⊠ collector ⊠ episcleral veins)
3. POAG: chronic painless, open angle, mild IOP; 1st line = PG analogs (latanoprost)
4. PACG: acute painful, narrow angle, IOP 40-80; Rx = IV Mannitol + Pilocarpine + Laser PI
5. Cup:Disc >0.8 or >0.2 asymmetry = glaucoma suspect
6. ISNT Rule: Inferior > Superior > Nasal > Temporal rim thickness; violation = suspect
7. Bjerrum (arcuate) scotoma = most characteristic field defect
8. Trabeculectomy = gold standard filtering surgery

POAG vs PACG

Feature	POAG	PACG
Onset	Insidious	Acute
Symptoms	Painless	Severe pain + vomiting
Angle	Open	Narrow/Closed
IOP	Mild (22-40)	Very high (40-80)
Pupil	Normal	Mid-dilated, fixed
Rx	PG analogs	Mannitol + Pilo + PI

ISNT rule violated?

Time to panic (professionally)

Memory Tricks!

ISNT Rule: Inferior thickest ⊠ Temporal thinnest

PACG emergency: Steamy cornea + Red eye + Fixed pupil + Severe pain

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q #1 irreversible blindness?	Glaucoma
Q Most common glaucoma?	POAG
Q Gold standard tonometry?	Goldmann Applanation (GAT)
Q Most characteristic field defect?	Bjerrum (arcuate) scotoma
Q Steamy cornea + fixed pupil?	Acute angle closure

DEFINITION

Inflammation of uveal tract (iris, ciliary body, choroid). Anterior uveitis (60-70%) is the most common type.

KEY FACTS

1. Granulomatous: mutton-fat KPs, Koeppe+Busacca nodules, chronic (TB, sarcoid, VKH)
2. Non-granulomatous: fine dusty KPs, acute (HLA-B27, JIA, idiopathic)
3. KPs settle in Arlt's triangle (inferior corneal endothelium)
4. Posterior synechiae ⊠ irregular pupil; 360° = seclusio pupillae
5. Complications: cataract, secondary glaucoma, band keratopathy, CME
6. Rx: Topical steroids + cycloplegics (atropine) for anterior; oral steroids for posterior
7. Rule out infection BEFORE starting steroids

Granulomatous vs Non-Granulomatous

Feature	Granulomatous	Non-Granulomatous
Onset	Insidious	Acute
KPs	Mutton-fat (large)	Fine, dusty, small
Iris nodules	Koeppe + Busacca	Absent
Course	Chronic relapsing	Acute self-limiting
Causes	TB, Sarcoid, Syphilis	HLA-B27, JIA

Mutton-fat KPs on exam

Granulomatous! TB until proven otherwise

Memory Tricks!

Seclusio = 360° ring synechiae; Occlusio = pupil membrane ⊠ iris bombe

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Most common uveitis?	Anterior (iridocyclitis)
Q Mutton-fat KPs?	Granulomatous (TB, sarcoid)
Q Stellate KPs?	Fuchs heterochromic iridocyclitis
Q Hypopyon uveitis?	Behcet disease
Q Band keratopathy in children?	JIA uveitis

Retinal Diseases

DEFINITION

Diseases of retina including diabetic retinopathy (DR), ARMD, vascular occlusions, and retinoblastoma.

KEY FACTS

1. DR: Mild NPDR (microaneurysms) ⊠ Severe NPDR (4-2-1 rule) ⊠ PDR (neovascularization)
2. 4-2-1 Rule: 4 quad hemorrhages + 2 quad venous beading + 1 quad IRMA = Severe NPDR
3. DME: Anti-VEGF first-line (ranibizumab, bevacizumab, aflibercept)
4. ARMD: Dry (90%, drusen) vs Wet (10%, CNVM, 90% severe vision loss)
5. CRAO: Cherry-red spot, EMERGENCY (90-min golden period), pale ischemic retina
6. CRVO: Blood & thunder fundus; ischemic ⊠ 100-day glaucoma (NVG)
7. RD: Rhegmatogenous (tear, MC) vs Tractional (PDR) vs Exudative (tumor)
8. Retinoblastoma: MC intraocular malignancy childhood; leukocoria = MC sign; RB1 gene 13q14

CRAO vs CRVO

Feature	CRAO	CRVO
Type	Arterial	Venous
Fundus	Cherry-red spot	Blood & thunder
Hemorrhages	Few/absent	Flame-shaped all 4 quad
Emergency?	YES (90 min)	Urgent referral
Complication	Optic atrophy	100-day NVG

Cherry red spot at macula

CRAO! We have 90 MINUTES. GO!

Memory Tricks!

4-2-1 = Severe NPDR ⊠ 50% progress to PDR in 1 year

Cherry red spot: CRAO + Tay-Sachs + Niemann-Pick

100-Day Glaucoma: Ischemic CRVO ⊠ NVG at ~100 days

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Cherry-red spot at macula?	CRAO
Q Blood & thunder fundus?	CRVO
Q 100-day glaucoma?	Ischemic CRVO ⊠ NVG
Q MC cause leukocoria?	Retinoblastoma
Q Anti-VEGF for DME?	Ranibizumab / Bevacizumab

Squint (Strabismus)

DEFINITION

Misalignment of visual axes; one eye deviates while other fixates. 2-4% of children.

KEY FACTS

1. Esotropia (inward) = MC in kids; Exotropia (outward) = MC in adults
2. Concomitant: same angle all gazes, childhood | Paralytic: varies with gaze, CN palsies
3. CN III palsy: eye down & out + ptosis + dilated pupil
4. CN IV: vertical diplopia, head tilt | CN VI: convergent, face turn
5. Accommodative esotropia: uncorrected hypermetropia ⊠ Rx: glasses first
6. Cover test: detects tropia | Hirschberg: corneal reflex (1mm = ~15 PD)
7. Amblyopia: reduced VA without organic cause; critical period <7-8 yrs
8. Treatment: patch GOOD eye (occlusion therapy), correct RE, surgery

Concomitant vs Paralytic

Feature	Concomitant	Paralytic
Angle	Same all directions	Varies with gaze
Age	Childhood	Any (often adult)
Diplopia	Usually absent	Present
Eye movements	Full	Restricted
Primary deviation	= Secondary	< Secondary

Patch the GOOD eye??

Yes. The GOOD one. Trust the process.

Memory Tricks!

LR6 SO4 Rest3

Parks 3-Step Test for CN IV palsy: hypertropic eye ⊠ worse gaze ⊠ worse tilt

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q MC squint in children?	Esotropia
Q Eye down & out + ptosis?	CN III palsy
Q Critical period amblyopia?	<7-8 years
Q Amblyopia treatment?	Patch GOOD eye
Q Screening test squint?	Hirschberg

Ocular Trauma

DEFINITION

BETT classification: Closed globe (contusion, lamellar) vs Open globe (rupture, penetrating, perforating, IOFB).

KEY FACTS

1. Hyphema grading: I (<1/3) to IV (total/8-ball); risk: re-bleed day 3-5, secondary glaucoma
2. Chemical burns: Alkali WORSE (liquefactive, deep) > Acid (coagulative, superficial)
3. First aid chemical burn: IMMEDIATE copious water irrigation 30+ min
4. Angle recession: ciliary muscle tear ⊠ late-onset OAG months-years later
5. Berlin edema (commotio retinae): posterior pole whitening, self-limiting
6. Siderosis (iron): heterochromia, b-wave gone on ERG | Chalcosis (copper): sunflower cataract
7. Roper-Hall grading: I (clear) to IV (opaque cornea, >1/2 limbal ischemia = poor)
8. MRI contraindicated in metallic IOFB

Siderosis vs Chalcosis

Feature	Siderosis (Iron)	Chalcosis (Copper)
Iris	Heterochromia (rust)	Green discoloration
Lens	Brown discoloration	Sunflower cataract
ERG	Extinguished (b-wave)	May be normal
Retina	Pigmentary degeneration	KF-like ring

Chemical burn? Which came first?

WATER. Always water. IMMEDIATELY.

Memory Tricks!

Alkali = liquefactive (deep, WORSE); Acid = coagulative (superficial, less bad)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q First aid chemical burn?	Immediate copious water irrigation
Q Alkali worse because?	Liquefactive necrosis (deep)
Q Rosette cataract?	Blunt trauma
Q Sunflower cataract?	Copper (chalcosis) / Wilson's
Q MRI contraindicated?	Metallic IOFB

Vitamin A Deficiency

DEFINITION

Spectrum: Night blindness (XN) ⊠ Bitot spots (X1B) ⊠ Corneal xerosis (X2) ⊠ Keratomalacia (X3B). WHO classification.

KEY FACTS

1. XN (night blindness) = earliest symptom, responds in 1-2 days
2. X1B (Bitot spots) = pathognomonic; foamy triangular, temporal conjunctiva; C. xerosis
3. X3B (keratomalacia) = EMERGENCY, corneal melting ⊠ permanent blindness
4. Risk: age 1-4yr, PEM, measles, diarrhea
5. WHO dose: <6mo 50,000 IU | 6-12mo 1,00,000 IU | >12mo 2,00,000 IU (Day 0, 1, 14)
6. India schedule: 1L IU at 9mo, 2L IU every 6mo until 5yr (9 total doses)
7. NEVER mega-dose pregnant women (teratogenic!)
8. 90% Vitamin A stored in liver; dark adaptation test = earliest functional test

WHO Classification of VAD

Stage	Sign	Key Detail
XN	Night blindness	Earliest, responds 1-2 days
X1B	Bitot spots	Pathognomonic, C. xerosis
X2	Corneal xerosis	Dull, hazy, reversible
X3A	Ulcer <1/3 cornea	Scarring likely
X3B	Keratomalacia ⊠1/3	EMERGENCY, melting

Bitot spots = pathognomonic

C. xerosis living rent-free on conjunctiva

Memory Tricks!

Bitot spots >0.5% OR Night blindness >1% in children = public health problem (WHO)

Never mega-dose pregnant women — teratogenic!

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Earliest symptom VAD?	Night blindness (XN)
Q Pathognomonic sign VAD?	Bitot spots (X1B)
Q Organism in Bitot spots?	Corynebacterium xerosis
Q Most severe ocular stage?	Keratomalacia (X3B)
Q Vit A dose >1yr?	2,00,000 IU Day 0, 1, 14

Blindness Prevention

DEFINITION

WHO blindness: VA <3/60 in better eye. 80% of global blindness is avoidable (preventable + treatable).

KEY FACTS

1. Causes India: Cataract 66.2% ▪ Corneal opacity 8.2% ▪ RE 7.2% ▪ Glaucoma 5.5%
2. Blindness prevalence India: 1.49% (1986) ▪ 0.36% (2015-19)
3. NPCBVI launched 1976 (India = first country); renamed 2017; target 0.3%
4. CSR ~6000/million/yr (India = highest globally); ~6.5M surgeries/yr
5. Vision 2020 (WHO+IAPB 1999): eliminate avoidable blindness; 5 priority diseases
6. SAFE strategy for trachoma: Surgery + Antibiotics + Face washing + Environment
7. Age >50: Cataract #1 | Age <50: Corneal opacity #1

Blindness Key Data

Parameter	Value
#1 cause India	Cataract (66.2%)
Prevalence India	0.36% (2015-19)
NPCB year	1976 (first country)
CSR target	6000/million/yr
Vision 2020	WHO+IAPB, 1999
Avoidable blindness	80% of total

80% blindness is avoidable!

Cataract surgeons: we got this

Memory Tricks!

SAFE = Surgery, Antibiotics, Face washing, Environment (trachoma)

Avoidable = Preventable + Treatable (80%)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q #1 cause blindness India?	Cataract (66.2%)
Q WHO blindness definition?	VA <3/60 better eye
Q NPCB year?	1976
Q Vision 2020 by?	WHO + IAPB (1999)
Q India CSR?	~6000/million/year

Ear: Hearing, Otitis + Deafness

DEFINITION

CSOM = chronic middle ear inflammation with permanent TM perforation >3 months. Tubotympanic (safe) vs Atticoantral (unsafe).

KEY FACTS

1. Safe (tubotympanic): central perforation, mucoid discharge, NO cholesteatoma, conservative Rx
2. Unsafe (atticoantral): marginal perforation, foul discharge, CHOLESTEATOMA (hallmark), ALWAYS surgery
3. Rinne: AC > BC = positive (normal/SNHL); BC > AC = negative (CHL)
4. Weber: lateralizes to diseased ear in CHL; lateralizes to BETTER ear in SNHL
5. Cholesteatoma = keratinized squamous epithelium sac (NOT a tumor); bone erosion risk
6. IC complications: meningitis (MC IC), brain abscess, lateral sinus thrombosis
7. EC complications: mastoid abscess, facial nerve palsy, labyrinthine fistula
8. Gradenigo triad: otorrhea + retro-orbital pain + CN VI palsy

Safe vs Unsafe CSOM

Feature	Safe (Tubotympanic)	Unsafe (Atticoantral)
Perforation	Central	Marginal/Attic
Discharge	Mucoid, non-foul	Foul, scanty
Cholesteatoma	Absent	PRESENT (hallmark)
Treatment	Conservative	ALWAYS surgical
Complications	Rare	Dangerous (IC)

Unsafe CSOM = cholesteatoma

It's not a tumor… but treat it like one

Memory Tricks!

Ototoxic PLAN C: Platinum, Loop diuretics, Aminoglycosides, NSAIDs, Chloroquine

False Negative Rinne: severe unilateral SNHL ⊠ cross-hearing from normal ear

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Unsafe CSOM hallmark?	Cholesteatoma
Q MC IC complication CSOM?	Meningitis
Q Gradenigo triad?	Otorrhea + orbital pain + CN VI
Q Rinne negative =?	Conductive hearing loss
Q Weber ⊠ affected ear =?	Conductive loss in that ear

Nose: Sinusitis, DNS + Epistaxis

DEFINITION

DNS = deviation of nasal septum from midline. Epistaxis from Little's area (Kiesselbach's plexus) = 90% of nosebleeds.

KEY FACTS

1. DNS = MC cause nasal obstruction; Rx: Septoplasty (procedure of choice)
2. Little's area: anteroinferior septum, 4 arteries anastomose, 90% of epistaxis
3. Anterior epistaxis: young, mild, Little's area | Posterior: elderly, severe, Woodruff's plexus
4. Sinusitis: Acute (<12wk, S. pneumoniae MC) vs Chronic (>12wk, allergy/polyps/fungi)
5. Maxillary = largest sinus, most common involved
6. FESS = gold standard chronic sinusitis; key: uncinate process removal first
7. Chandler's grading: orbital complications of sinusitis (preseptal cellulitis = MC)
8. Atrophic rhinitis (ozaena): Klebsiella ozaenae, crusts, wide nasal cavity

Anterior vs Posterior Epistaxis

Feature	Anterior	Posterior
Site	Little's (Kiesselbach)	Woodruff's plexus
Age	Young	Elderly
Frequency	90%	10%
Severity	Mild, self-limiting	Severe
Artery	Anterior ethmoidal	Sphenopalatine

90% epistaxis is from Little's area

Little area, BIG bleed energy

Memory Tricks!

Cottle's test: pull cheek ⊠ improved breathing = nasal valve obstruction

OMC = Ostiomeatal Complex = drainage pathway for maxillary, anterior ethmoid, frontal

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q MC cause nasal obstruction?	DNS
Q MC site epistaxis?	Little's area (Kiesselbach)
Q MC organism acute sinusitis?	S. pneumoniae
Q MC sinus involved?	Maxillary
Q Atrophic rhinitis organism?	Klebsiella ozaenae

Throat: Tonsillitis + Larynx

DEFINITION

Acute tonsillitis = Group A beta-hemolytic Streptococcus. Laryngeal CA = SCC >95%, smoking #1 risk.

KEY FACTS

1. Quinsy (peritonsillar abscess): unilateral sore throat, trismus, hot potato voice; Rx = I&D + IV Abx
2. Paradise 7-5-3: 7 episodes/yr OR 5/yr x2yr OR 3/yr x3yr = tonsillectomy indication
3. Adenoid facies: open mouth, narrow palate, pinched nose; OSA risk
4. Semon's Law: Abductors affected FIRST in progressive RLN lesion ⊠ median ⊠ cadaveric position
5. Unilateral RLN palsy: hoarse breathy voice | Bilateral: stridor, near-normal voice
6. Glottic CA (60%): best prognosis, early hoarseness, rare mets
7. Supraglottic CA (35%): common LN mets, worse prognosis

Glottic vs Supraglottic CA

Feature	Glottic (60%)	Supraglottic (35%)
Early symptom	Hoarseness (EARLY)	Dysphagia
LN mets	RARE	COMMON
Prognosis	BEST	Intermediate
Rx	RT/cordectomy	RT +/- surgery

Paradise criteria: 7-5-3

Me counting my own sore throats

Memory Tricks!

Paradise 7-5-3 Rule = tonsillectomy indication

Semon's Law: Abductors FIRST ⊠ cord median ⊠ then cadaveric

'To Zanzibar By Motor Car' = 5 branches facial nerve (Temporal, Zygomatic, Buccal, Marginal mandibular, Cervical)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q MC organism tonsillitis?	Group A Strep
Q Hot potato voice + trismus?	Quinsy (peritonsillar abscess)
Q Paradise criteria?	7-5-3 (tonsillectomy indication)
Q Adenoid facies?	Open mouth + narrow palate
Q Semon's law?	Abductors affected first (RLN)

Vertigo

DEFINITION

Illusion of movement (rotation). 90% peripheral (inner ear), 10% central (brain). NOT just dizziness!

KEY FACTS

1. BPPV = #1 cause; otoconia in posterior SCC; episodes <1 min; positional
2. Dix-Hallpike = diagnostic gold standard for BPPV; Epley maneuver = Rx (80% cure)
3. Meniere's triad: Episodic vertigo + Tinnitus + Fluctuating SNHL (+ aural fullness = tetrad)
4. Meniere's pathology: endolymphatic hydrops; Rx = low-salt + diuretics + betahistine
5. Vestibular neuritis: acute vertigo + NO hearing loss | Labyrinthitis: + hearing loss
6. COWS = Cold Opposite, Warm Same (fast phase direction in caloric test)
7. Canal paresis >20-25% on caloric = unilateral vestibular hypofunction
8. Nystagmus: peripheral = horizontal/torsional, fatigable; central = vertical, non-fatigable

Peripheral vs Central Vertigo

Feature	Peripheral (90%)	Central (10%)
Onset	Sudden, severe	Gradual/sudden, mild
Nystagmus	Horizontal/torsional	Vertical/multidirectional
Fixation	Suppressed	NOT suppressed
Hearing loss	Common	Rare
Neuro signs	Absent	Present
Fatigability	Fatigues	Does NOT fatigue

Room is spinning after studying

Is it BPPV or just finals week?

Memory Tricks!

COWS = Cold Opposite, Warm Same (caloric fast phase)

BPPV nystagmus: Latent, Torsional, Fatigable, Brief (<60s) = PERIPHERAL

Meniere's = ELF: Episodic vertigo + Low-freq HL + Fullness

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q MC cause vertigo?	BPPV
Q Gold standard BPPV test?	Dix-Hallpike
Q BPPV treatment?	Epley maneuver
Q Meniere's triad?	Vertigo + tinnitus + SNHL
Q COWS stands for?	Cold Opposite, Warm Same

Facial Nerve Palsy

DEFINITION

CN VII weakness/paralysis. UMN = forehead SPARED (bilateral cortical supply). LMN = entire face ipsilateral.

KEY FACTS

1. Bell's palsy (72%): MC cause, rapid onset, idiopathic (HSV-1), 85% recover; Rx = prednisolone <72hrs
2. Ramsay Hunt: VZV reactivation; triad = LMN palsy + ear vesicles + otalgia; 50-60% recovery
3. UMN: forehead SPARED (can wrinkle), lower face contralateral (stroke, tumor)
4. LMN: ENTIRE face ipsilateral, lagophthalmos, can't wrinkle forehead (Bell's, Hunt, otitis)
5. Labyrinthine segment = NARROWEST (most vulnerable to swelling)
6. House-Brackmann grading I-VI (I=normal, VI=total paralysis)
7. Topographic diagnosis: taste loss (above chorda), hyperacusis (above stapedius), tear loss (above petrosal)
8. 5 terminal branches: Temporal, Zygomatic, Buccal, Marginal mandibular, Cervical

Bell's vs Ramsay Hunt

Feature	Bell's Palsy	Ramsay Hunt
Cause	HSV-1 (likely)	VZV (Herpes Zoster)
Vesicles	ABSENT	PRESENT (ear)
Hearing loss	Usually absent	SNHL may occur
Pain	Mild post-auricular	Severe otalgia
Recovery	Good (85%)	Poorer (50-60%)

Forehead spared?

UMN! That's the whole trick.

Memory Tricks!

Forehead Sparing = UMN (bilateral cortical innervation of upper face)

'To Zanzibar By Motor Car' = T, Z, B, Mm, C (5 branches)

Labyrinthine = NARROWEST segment (most susceptible to compression)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q MC cause facial palsy?	Bell's palsy (72%)
Q Forehead spared?	UMN lesion
Q Vesicles on ear + palsy?	Ramsay Hunt (VZV)
Q Narrowest facial canal segment?	Labyrinthine
Q Hyperacusis damage?	Nerve to stapedius

Tumors of ENT

DEFINITION

ENT tumors: NPC (EBV), JNA (adolescent males), Laryngeal CA (SCC >95%), Acoustic neuroma (CPA tumor).

KEY FACTS

1. NPC: undifferentiated SCC, EBV, fossa of Rosenmuller; Trotter's triad (deafness+palate+CN V)
2. JNA: adolescent males ONLY, sphenopalatine origin, NEVER biopsy; pre-op embolization + excision
3. Laryngeal CA: Glottic 60% (best prognosis) > Supraglottic 35% > Subglottic 5% (worst)
4. Acoustic neuroma: benign CN VIII, MC CPA tumor; bilateral = NF2; MRI gold standard
5. Inverted papilloma: benign, aggressive, high recurrence, may turn malignant
6. Ohngren's line: medial canthus ⊠ mandible angle; superoposterior = worse prognosis
7. Pleomorphic adenoma = MC benign parotid; Mucoepidermoid CA = MC malignant salivary

Laryngeal CA by Site

Site	Frequency	Prognosis	Early Symptom
Glottic	60%	BEST	Hoarseness (early)
Supraglottic	35%	Intermediate	Dysphagia
Subglottic	5%	WORST	Stridor (late)

Adolescent male + nosebleed

JNA! And DO NOT biopsy!!

Memory Tricks!

JNA = Adolescent Male + Epistaxis ⊠ NEVER biopsy (torrential bleeding)

Glottic CA = BEST prognosis (early hoarseness + minimal lymphatics)

Ohngren's line: superior = worse; inferior = better

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Adolescent male + epistaxis + mass?	JNA (never biopsy!)
Q EBV nasopharyngeal tumor?	NPC
Q MC CPA tumor?	Acoustic neuroma
Q Bilateral acoustic neuromas?	NF2
Q MC laryngeal CA site?	Glottis (60%)

Airway Emergencies

DEFINITION

Life-threatening airway obstruction requiring immediate intervention. Stridor, FB aspiration, emergency airway procedures.

KEY FACTS

1. Stridor types: Inspiratory (supraglottic) | Expiratory (tracheobronchial) | Biphasic (subglottic)
2. Croup: parainfluenza, 6mo-3yr, barking cough, steeple sign; Rx = nebulized adrenaline + steroids
3. Epiglottitis: H. influenzae b, 2-7yr, high fever, drooling, NO cough, thumb sign; Rx = IV Abx + airway
4. FB aspiration: <3yr, right main bronchus MC (wider, more vertical); Rx = rigid bronchoscopy
5. Cafe coronary: food bolus impaction in larynx; Rx = Heimlich; infants = back blows + chest thrusts
6. Tracheostomy: 2nd-4th rings (NEVER 1st = stenosis); most fatal complication = tracheo-innominate fistula
7. Cricothyrotomy: emergency CICV; cricothyroid membrane; contraindicated <12yr
8. Ludwig's angina: floor-of-mouth cellulitis, dental origin, woody swelling; SECURE AIRWAY FIRST

Croup vs Epiglottitis

Feature	Croup	Epiglottitis
Organism	Parainfluenza	H. influenzae b
Age	6mo-3yr	2-7yr
Onset	Gradual	Rapid
Cough	Barking	Absent
Drooling	Absent	Present
X-ray	Steeple sign	Thumb sign

Steeple sign on X-ray

Croup! Not epiglottitis! (whew)

Memory Tricks!

Steeple = Croup; Thumb = Epiglottitis

FB aspiration ⊠ Right bronchus (wider, shorter, more vertical)

Ludwig's 4 Ds: Dental, Dusky, Doughy, Dyspnea

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q MC neonatal stridor?	Laryngomalacia
Q MC acute stridor children?	Croup
Q Steeple sign?	Croup
Q Thumb sign?	Epiglottitis
Q FB aspiration MC site?	Right main bronchus

Legal Procedures

DEFINITION

Inquest = inquiry into cause of unnatural, sudden, or suspicious death. Police (CrPC 174) vs Magistrate (CrPC 176).

KEY FACTS

1. Police Inquest (174): external exam only, NO dissection, all unnatural deaths
2. Magistrate Inquest (176): custody + dowry deaths <7yr; can order full autopsy
3. Dying Declaration (Sec 32 IEA): valid if compos mentis, NO oath needed, oral/written/gestures
4. Consent: Implied, Express, Informed, Proxy, Blanket (invalid), Loco parentis
5. Medical Negligence 4 Ds: Duty, Dereliction, Direct cause, Damage
6. Bolam Test (1957): NOT negligent if acting per responsible body of medical opinion
7. CPA 2019: Medical services = 'service'; free govt treatment NOT covered
8. Emergency consent: Sec 92 IPC (good faith, no consent needed)

Police vs Magistrate Inquest

Feature	Police (174)	Magistrate (176)
Authority	SHO (police officer)	Magistrate (judicial)
Cases	All unnatural deaths	Custody + dowry <7yr
Body exam	External only	Full autopsy possible
Report to	Magistrate	State government

174 or 176? THINK FAST!

Police = 174, Magistrate = 176. DONE.

Memory Tricks!

4 Ds of Negligence: Duty, Dereliction, Direct cause, Damage

Police = 174 (external); Magistrate = 176 (custody/dowry)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Police inquest section?	CrPC 174
Q Bolam test year?	1957 (UK)
Q Jacob Mathew case?	2005 (Indian negligence landmark)
Q Consent age surgery?	12 years (IPC)
Q Emergency consent?	Sec 92 IPC (good faith)

IPC + CrPC Basics

DEFINITION

IPC = Indian Penal Code (defines crimes). CrPC = Criminal Procedure Code (governs trials). BNS/BNSS replaced in 2024.

KEY FACTS

1. Murder (302): Intention to cause death | Culpable Homicide (299): Knowledge death MAY occur
2. All murder IS culpable homicide, but NOT all CH is murder
3. Grievous Hurt (8 types): Emasculation, Eye, Ear, Privation, Disfigure, Fracture, Endanger life
4. Rape (375/BNS 63): against will, <18yr = statutory; Min 10yr RI to life
5. Insanity (84): McNaughten Rule (1843) — no liability if unsound mind
6. Death by Negligence (304A): 2yr jail
7. Suicide attempt (309): DECRIMINALIZED by MHCA 2017
8. BNS/BNSS July 2024: Murder 302→103, CH 299→100, Rape 375→63

Murder vs Culpable Homicide

Feature	Murder (302)	CH not murder (299)
Intention	To cause death	Knowledge death likely
Degree	Highest	Lesser
Punishment	Life / Death	10yr / Life
BNS	Section 103	Section 100

All murder is CH...

But not all CH is murder *mind blown*

Memory Tricks!

All murder is CH, but all CH is NOT murder (golden rule)

McNaughten Rule (1843): nature of act + wrong = insanity defense

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Murder section?	IPC 302 / BNS 103
Q Rape section?	IPC 375 / BNS 63
Q Insanity defense?	IPC 84 (McNaughten 1843)
Q Death by negligence?	304A (2yr jail)
Q Suicide attempt decriminalized?	MHCA 2017

Medical Ethics

DEFINITION

Ethical principles governing medical practice. NMC (2020) replaced MCI. Brain death = legal death.

KEY FACTS

1. NMC 2020: 4 Autonomous Boards (UGMEB, PGMEB, MARB, EMRB)
2. Euthanasia: Active (illegal), Passive (legal — Common Cause v UoI 2018); Aruna Shanbaug 2011
3. Brain Death: absent brainstem reflexes + apnea test + flat EEG (2x, 6hr gap) = LEGAL death
4. THOA 1994: Brain death committee = 4 doctors; organ sale = 5yr jail + Rs 20L fine
5. Professional Misconduct: adultery with patient, false certificates, fee-splitting, self-advertising
6. Confidentiality exceptions: court order, notifiable diseases, gunshot/stab wounds
7. Medical records: maintain minimum 3 years
8. Helsinki Declaration (1964), Nuremberg Code (1947): clinical trial ethics

Types of Euthanasia

Type	Description	Legal Status (India)
Active	Deliberate act to end life	Illegal
Passive	Withdraw life support	Legal (2018)
Voluntary	Patient's own wish	Advance directive legal
Involuntary	Without patient consent	Illegal

Brain death = legal death

4 doctors agree: no coming back

Memory Tricks!

Brain Death = Legal Death! 4 doctors certify, brainstem death irreversible

THOA 1994: organ sale = 5yr + Rs 20 lakh fine

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Passive euthanasia legalized?	Common Cause v UoI (2018)
Q First euthanasia case India?	Aruna Shanbaug (2011)
Q Brain death committee?	4 doctors
Q MCI replaced by?	NMC (2020)
Q Organ sale penalty?	5yr jail + Rs 20L fine

DEFINITION

Corpus Delicti = proving the body of the crime. Methods: fingerprints (routine), DNA (gold standard), dental, skeletal.

KEY FACTS

1. Fingerprints: appear 12-16wk IU, permanent, unique (not even identical twins)
2. Galton's: Loop 67% (MC), Whorl 25%, Arch 7%, Composite 1% (rarest)
3. DNA profiling: Sir Alec Jeffreys (1984 UK); unique except identical twins
4. Gustafson's 6 parameters: Attrition, Periodontosis, Secondary dentin, Cementum, Resorption, Transparency
5. Skull-photo superimposition: 16 landmarks, first in Ruxton case (1935)
6. Teeth = most resistant body structure; dental ID very reliable
7. Age estimation: teeth, ossification, secondary sexual characters, radiological

Fingerprint Types (Galton)

Type	Frequency	Pattern
Loop	67% (MC)	One delta, recurving ridge
Whorl	25%	Two deltas, circular
Arch	7%	No delta, wave-like
Composite	1% (rarest)	Mixed patterns

67% of fingerprints are loops

The other 33% chose to be difficult

Memory Tricks!

APSCRT: Gustafson's 6 = Attrition, Periodontosis, Sec dentin, Cementum, Resorption, Transparency

Fingerprints = most reliable routine; DNA = gold standard (expensive)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Most reliable routine ID?	Fingerprints
Q MC fingerprint type?	Loop (67%)
Q DNA profiling pioneer?	Sir Alec Jeffreys (1984)
Q Most resistant structure?	Teeth
Q Gustafson parameters?	6 (APSCRT)

Postmortem Changes

DEFINITION

Physical & chemical changes after death: cooling (algor), lividity (livor), stiffening (rigor), decomposition.

KEY FACTS

1. Algor mortis: ~1.5°F/hr cooling; Henssge nomogram most accurate
2. Livor mortis: 30min-1hr onset, 6-12hrs FIXED (no blanching on pressure)
3. Rigor mortis: 1-2hr start, 12hr full, 24hr disappears; Nysten's Rule (eyelids⊠jaw⊠neck⊠trunk⊠limbs)
4. Putrefaction: 24-36hr green RIF (first external sign), 48-72hr marbling, 3-5d bloating
5. First internal organ to putrefy: Larynx & Trachea
6. Last organ: Non-gravid uterus (female) / Prostate (male)
7. Adipocere: warm+moist+anaerobic, C. welchii, 3wk-6mo | Mummification: hot+dry, 3mo-1yr
8. Casper's Dictum: Air:Water:Earth = 1:2:8 decomposition ratio

PM Changes Timeline

Change	Onset	Key Detail
Algor mortis	Immediate	~1.5°F/hr cooling
Livor mortis	30min-1hr	Fixed at 6-12hr
Rigor mortis	1-2hr	Full 12hr, gone 24hr
Putrefaction	24-36hr	Green RIF first
Adipocere	3wk-6mo	Warm + moist
Mummification	3mo-1yr	Hot + dry

Green discoloration where?

Right iliac fossa! ALWAYS the RIF!

Memory Tricks!

Nysten's Rule: Eyelids ⊠ Jaw ⊠ Neck ⊠ Trunk ⊠ Upper ⊠ Lower limbs

Casper's Dictum: 1:2:8 (Air:Water:Earth)

Cherry-red lividity = CO poisoning

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q First external putrefaction sign?	Green discoloration, RIF
Q First internal organ to putrefy?	Larynx & Trachea
Q Cadaveric spasm =?	Instantaneous rigor (no flaccidity)
Q Cherry-red lividity =?	CO poisoning
Q Casper's ratio?	1:2:8 (air:water:earth)

Injuries: Types + Weapons

DEFINITION

Injury (Sec 44 IPC) = any harm illegally caused to body, mind, reputation, or property.

KEY FACTS

1. Abrasion: skin scraped, minimal bleeding, heals without scar
2. Contusion: bleeding under intact skin, color changes (red⊠blue⊠green⊠yellow 14 days)
3. Laceration: torn by blunt force, ragged edges, hair follicles INTACT (crushed, not cut)
4. Incised: sharp edges, hair CLEANLY CUT, profuse bleeding, fine scar
5. Stab wound: depth > surface length; single-edged = one sharp + one blunt angle
6. Firearm: Entry (round, inverted, collar of abrasion) vs Exit (irregular, larger, everted)
7. Burns Rule of 9: Head 9%, Arms 9% each, Trunk 18% each, Legs 18% each, Perineum 1%
8. AM burns: vital reaction, soot in airways, CO-Hb >10%

Laceration vs Incised Wound

Feature	Laceration (Blunt)	Incised (Sharp)
Edges	Irregular, ragged	Clean, sharp
Hair	Intact (crushed)	Cleanly cut
Bleeding	Less (crushed vessels)	Profuse
Healing	Irregular scar	Fine linear scar
Bridge tissue	Present	Absent

Hair intact and edges ragged?

LACERATION! Not incised! I know this!

Memory Tricks!

Hair intact + ragged = LACERATION; Hair cut + sharp = INCISED

Collar of abrasion = ENTRY wound (firearm)

Contusion colors: 14 days cycle (red⊠blue⊠green⊠yellow⊠brown)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Hair intact, ragged edges?	Laceration (blunt)
Q Stab wound feature?	Depth > surface length
Q Tattooing in firearm?	Close range (<60cm)
Q Rule of 9 perineum?	1% TBSA
Q Collar of abrasion?	Entry wound (firearm)

Asphyxia

DEFINITION

Condition caused by interference with respiration or lack of oxygen. Types: hanging, strangulation, suffocation, drowning.

KEY FACTS

1. Tardieu spots: petechial hemorrhages on pleura/pericardium (from capillary rupture)
2. Hanging: body weight, oblique mark ABOVE thyroid, pale face, MC SUICIDAL
3. Strangulation: other person force, horizontal mark AT/BELOW thyroid, congested face, MC HOMICIDAL
4. Throttling: crescentic nail marks + finger-tip bruises on neck (pathognomonic)
5. C2 fracture (hangman's) in judicial hanging (long drop)
6. Drowning: Wet (90%), Dry (laryngospasm); Fresh water (hemodilution, VF) vs Salt (hemoconcentration)
7. Diatom test: unicellular algae in bone marrow = ante-mortem drowning
8. Hyoid fracture more common in strangulation (age >40yr)

Hanging vs Ligature Strangulation

Feature	Hanging	Strangulation
Force	Body weight	Other person
Ligature mark	Oblique, above thyroid	Horizontal, at/below
Face	Pale (typical)	Congested
Manner	Suicidal (MC)	Homicidal (MC)
Knot	Above ear (typical)	At back of neck

Hanging = suicidal (usually)

Strangulation = homicidal. Classic trap Q.

Memory Tricks!

Hanging = Suicidal (MC); Strangulation = Homicidal (MC)

Diatom test = proof of ante-mortem drowning (in bone marrow/organs)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q MC manner of hanging?	Suicidal
Q MC manner of strangulation?	Homicidal
Q Ligature mark strangulation?	Horizontal, at/below thyroid
Q Classic drowning sign?	Fine white froth at mouth/nose
Q Hyoid fracture MC in?	Strangulation (>40yr)

Poisoning

DEFINITION

Any substance that when introduced to body can injure health or destroy life. OP poisoning = most common in India.

KEY FACTS

1. OP mechanism: inhibits AChE irreversibly ⊠ ACh accumulation
2. DUMBELS (muscarinic): Diarrhea, Urination, Miosis, Bradycardia, Emesis, Lacrimation, Salivation
3. OP Rx: Atropine (muscarinic block) + PAM/Pralidoxime (reactivates AChE within 24hr)
4. CO poisoning: cherry-red lividity, bright pink blood; COHb >60% = death
5. Methanol: blindness (retina target) | Ethanol: inebriation; Methanol Rx = ethanol/fomepizole
6. Arsenic: ideal homicidal poison (tasteless), rice-water stools, garlic odor; Rx = BAL
7. Lead: Burton's line, wrist drop, basophilic stippling, colic
8. Snake Big 4 India: Cobra + Krait (neurotoxic), Russell's Viper + Saw-scaled (vasculotoxic)

Common Antidotes

Poison	Antidote
OP/Carbamate	Atropine + PAM
Opioids	Naloxone
Benzodiazepines	Flumazenil
Cyanide	Nitrite + Thiosulfate
Methanol	Ethanol / Fomepizole
Arsenic/Mercury	BAL (Dimercaprol)

DUMBELS = OP effects

Diarrhea Urination Miosis Bradycardia...

Memory Tricks!

DUMBELS = OP muscarinic effects

Queen of Poisons = Aconite | King = Arsenic

Dhatura: 'Hot as Hare, Blind as Bat, Dry as Bone, Red as Beet, Mad as Hatter'

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Ideal homicidal poison?	Arsenic
Q Cherry-red lividity?	CO poisoning
Q Pin-point pupils?	OP / Opioid poisoning
Q Rice-water stools poison?	Arsenic
Q Snowstorm vision?	Methanol

Sexual Offences

DEFINITION

Rape (IPC 375/BNS 63): sexual intercourse against will/consent. Age of consent = 18 years.

KEY FACTS

1. Elements: against will, fear of death/hurt, misconception, intoxication, <18yr (statutory)
2. Victim exam: female doctor + female attendant; general + genital exam + samples
3. Accused exam (CrPC 53): can examine WITHOUT consent; potency check + samples
4. Intact hymen does NOT rule out rape; Absence of sperm does NOT rule out rape
5. POCSO Act 2012: children <18yr, BOTH sexes, gender-neutral, mandatory reporting
6. Hymen types: Annular (MC), Crescentic, Fimbriated, Septate, Cribriform
7. Semen tests: Florence (screening) ⊠ PSA (confirmatory) ⊠ DNA (definitive)
8. Punishment: general 10yr-life; gang 20yr-life; <12yr 20yr-death

Semen Detection Tests

Test	Type	Detects
Florence	Screening	Choline crystals
Barberio	Screening	Spermine crystals
Acid Phosphatase/PSA	Confirmatory	Prostatic secretion
DNA profiling	Definitive	Individual ID

Intact hymen = no rape?

WRONG! That's the biggest myth in FM!

Memory Tricks!

Florence = First (screening) | PSA = Proof | DNA = Definitive

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Rape section?	IPC 375 / BNS 63
Q Age of consent?	18 years
Q POCSO year?	2012
Q MC hymen type?	Annular
Q Semen confirmatory?	PSA (p30)

Age Estimation

DEFINITION

Determining age from biological markers. Methods: dentition, ossification, secondary sexual characters, Gustafson's.

KEY FACTS

1. First deciduous tooth: lower central incisor (6mo); all 20 by 2.5yr
2. First permanent tooth: 1st molar (6yr); wisdom tooth 17-25yr (most variable)
3. First ossification center: lower femur (36wk IU)
4. Last epiphysis to fuse: medial clavicle (25yr) = marks skeletal maturity
5. Gustafson's 6: Attrition, Periodontosis, Secondary dentin, Cementum, Resorption, Transparency
6. Best single site radiological: WRIST
7. Criminal responsibility: <7yr = doli incapax (no liability); 7-12yr = immature understanding
8. Legal ages: 7 (criminal), 12 (consent), 18 (majority/consent/rape), 21 (male marriage)

Key Age Milestones

Milestone	Age	Significance
1st deciduous tooth	6 months	Lower central incisor
1st permanent tooth	6 years	1st molar
1st ossification	36 wk IU	Lower femur
Last epiphysis	25 years	Medial clavicle
Criminal liability	7 years	Doli incapax below
Legal majority	18 years	Consent, vote, marry (F)

First ossification center?

Lower femur at 36 weeks. LOCKED IN.

Memory Tricks!

FIRST ossification = Lower femur (36wk IU); LAST epiphysis = Medial clavicle (25yr)

Gustafson = APSCRT (6 dental parameters for adult age)

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q First deciduous tooth?	Lower central incisor (6mo)
Q First permanent tooth?	1st molar (6yr)
Q First ossification center?	Lower femur (36wk)
Q Last epiphysis to fuse?	Medial clavicle (25yr)
Q Doli incapax?	Below 7 years

Case Taking + History

DEFINITION

Systematic documentation of patient's medical history. History = 80% of diagnosis!

KEY FACTS

1. Format: Biodata ⊠ C/C ⊠ HOPI ⊠ Past Hx ⊠ Personal ⊠ Family ⊠ Menstrual ⊠ Rx History
2. Chief complaints: patient's OWN words, chronological, with duration, max 3-4
3. SOCRATES: Site, Onset, Character, Radiation, Associations, Timing, Exacerbating, Severity
4. Past Hx: DM, HTN, TB, asthma, surgeries, drugs, allergies, transfusions
5. Personal: diet, sleep, bowel/bladder, habits (pack-years, units), occupation, marital
6. Smoking pack-years = (packs/day) x years; Alcohol units = (vol x %) / 1000
7. Do: introduce yourself, stand on RIGHT side, open-ended questions, eye contact
8. Don't: jargon with patient, interrupt, leading questions, write diagnosis before exam

SOCRATES Framework

Letter	Component	Ask About
S	Site	Where exactly?
O	Onset	Sudden / gradual? When?
C	Character	Sharp / dull / burning?
R	Radiation	Does it spread?
A	Associations	Other symptoms with it?
T	Timing	Constant / intermittent?
E	Exacerbating	What makes it worse/better?
S	Severity	Scale 1-10?

History = 80% of diagnosis

The other 20% is vibes (jk, it's examination)

Memory Tricks!

SOCRATES = complete pain history framework

80% of diagnoses come from HISTORY alone!

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Most important for diagnosis?	History (80%)
Q C/C recorded in?	Patient's own words + duration
Q Stand on which side?	Right side
Q Menstrual Hx must include?	LMP, cycle, regularity, flow
Q SOCRATES for?	Pain history framework

DEFINITION

Systematic physical examination: Inspection ⊠ Palpation ⊠ Percussion ⊠ Auscultation.

KEY FACTS

1. Vitals: Pulse 60-100, BP 120/80, Temp 37°C, RR 12-20, SpO2 >95%
2. General exam: PICICLEJ = Pallor, Icterus, Cyanosis, Clubbing, Lymphadenopathy, Edema, JVP
3. Clubbing: angle >180°, Schamroth sign (obliterated diamond window)
4. Visual acuity: Snellen (6/6), FC, HM, PL; Pinhole test for refractive error
5. Pupil: 3-5mm, direct reflex, consensual, RAPD (Marcus Gunn), anisocoria
6. Rinne: AC > BC = positive (normal/SNHL); BC > AC = negative (CHL)
7. Weber: lateralizes to NORMAL ear in SNHL; lateralizes to AFFECTED in CHL
8. Normal IOP: 10-21 mmHg

Rinne & Weber Interpretation

Scenario	Rinne	Weber
Normal	Positive (AC>BC)	Central (no lateralization)
Conductive HL	Negative (BC>AC)	To affected ear
Sensorineural HL	Positive (AC>BC)	To normal ear

PICICLEJ on every patient

Pallor Icterus Cyanosis... *checks notes*

Memory Tricks!

PICICLEJ: Pallor, Icterus, Cyanosis, Clubbing, LN, Edema, JVP

Rinne: AC > BC = positive = NORMAL; BC > AC = negative = CONDUCTIVE LOSS

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Normal VA?	6/6 (Snellen)
Q Normal IOP?	10-21 mmHg
Q Negative Rinne?	Conductive hearing loss
Q Weber to normal ear?	Sensorineural loss
Q Schamroth sign?	Clubbing

Community Visits + Field Surveys

DEFINITION

Systematic door-to-door survey assessing family health, morbidity, and determinants of community health.

KEY FACTS

1. Steps: Planning ⊠ Sampling ⊠ Tool Prep ⊠ Field Visit ⊠ Data Entry ⊠ Analysis ⊠ Report ⊠ Feedback
2. Community diagnosis = clinical diagnosis for a POPULATION
3. SES scales: Kuppuswamy (EOI: Education+Occupation+Income, urban) vs BG Prasad (income, simpler)
4. Sample size: $n = Z^2 \times P(1-P) / d^2$; Z=1.96 (95%CI); add 10-20% for non-response
5. Sampling: Simple random, Systematic (every kth), Stratified, Cluster, Multistage
6. 30x7 Cluster sampling: WHO standard for immunization coverage
7. Data collection: interview, observation, record review, FGD, anthropometry

Sampling Methods

Method	Description	Best For
Simple Random	Every person equal chance	Small population
Systematic	Every kth person	Organized lists
Stratified	Divide then random	Heterogeneous groups
Cluster	Entire groups selected	Large geographic areas
Multistage	Multiple stages	National surveys

Sample size formula

n = Z-squared times... *cries in stats*

Memory Tricks!

Kuppuswamy = EOI (Education + Occupation + Income)

Z = 1.96 for 95% CI; Z = 2.58 for 99% CI

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q Family survey type?	Cross-sectional (descriptive)
Q Kuppuswamy parameters?	Education + Occupation + Income
Q Z for 95% CI?	1.96
Q Cluster sampling advantage?	Practical, cheaper
Q Non-response correction?	Add 10-20% to sample

Public Health Programs (Practical)

DEFINITION

Practical implementation of vaccination, nutrition screening, health education, and child health programs.

KEY FACTS

1. BCG: intradermal, left deltoid; OPV: oral; IPV: IM/SC; HepB: IM, birth within 24hrs
2. Cold chain: 2-8°C; ILR (2-8°C), Deep Freezer (-20°C)
3. VVM: color change shows heat exposure; Stage 1-2 USE, Stage 3-4 DISCARD
4. Freeze-sensitive (NEVER freeze): DPT, HepB, TT, IPV
5. Heat-sensitive (keep cold): BCG, Measles, OPV, JE
6. MUAC: >135mm GREEN | 125-135 YELLOW | 115-125 ORANGE (MAM) | <115 RED (SAM)
7. IMNCI: 0-5yr; danger signs = can't drink, vomiting, convulsions, lethargic
8. RBSK 4 Ds: Defects at birth, Diseases, Deficiencies, Development delays (0-18yr)

MUAC Classification

MUAC (mm)	Color	Status	Action
>135	GREEN	Normal	Routine monitoring
125-135	YELLOW	At risk	Supplementary feeding
115-125	ORANGE	MAM	Therapeutic feeding
<115	RED	SAM	Emergency referral

VVM stage 3-4?

DISCARD! Inner square darker = bad!

Memory Tricks!

VVM: 1-2 = USE; 3-4 = DISCARD (inner square darker = discard)

MUAC traffic light: GREEN ⊠ YELLOW ⊠ ORANGE ⊠ RED

RBSK = 4 Ds: Defects, Diseases, Deficiencies, Delays

RAPID FIRE — MCQ / Viva

EXAM FAV!

Q BCG route?	Intradermal (left deltoid)
Q Cold chain temp?	2-8°C
Q MUAC <115mm?	SAM (emergency)
Q Most heat-sensitive vaccine?	BCG
Q RBSK screens for?	4 Ds (Defects, Diseases, Deficiencies, Delays)